AI-Powered Thesis Management System: Design and Implementation of a Comprehensive Academic Planning Tool

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GitHub Repository: https://github.com/ananmouaz/thesis_agent

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Abstract

This paper describes an **AI-powered thesis management system** designed to help students plan, organize, and complete their academic research. The system combines **local AI models (Ollama)** and **cloud AI (Google Gemini)** with a web platform. It offers **automated timeline generation**, **task management**, and **18 specialized academic tools**. Built with a **Next.js/React frontend** and a **FastAPI backend**, it integrates with Notion for project management and Gmail for notifications. The system addresses common academic challenges by providing AI-assisted brainstorming, detailed task breakdowns, progress tracking, and ethical AI use monitoring.

Keywords: artificial intelligence, academic planning, thesis management, educational technology, web development

1. Introduction

Completing a thesis is often one of the hardest academic tasks. Students frequently struggle with managing literature, research methods, data, analysis, and writing over long periods, leading to stress and poor time management.

Recent AI advances, especially large language models (LLMs), offer new ways to help. AI can assist with many parts of academic work, from brainstorming topics to creating timelines and providing research tools. However, using AI in academics also raises important questions about integrity and how to balance AI help with independent work.

2. Materials & Methods

2.1 System Architecture

The Thesis Helper system uses a modern web application design. The **frontend** is built with **Next.js 14, React 18, and TypeScript** for a responsive user interface, styled with Tailwind CSS. The **backend** uses **FastAPI (Python)** with SQLAlchemy for database tasks and Pydantic for data checks. It uses SQLite for development and can use PostgreSQL for larger setups. Asynchronous programming helps handle AI and external calls efficiently.

2.2 AI Integration Framework

We use two main AI providers for flexibility:

- **Local AI (Ollama):** For privacy, Llama models run directly on the user's computer, keeping sensitive academic content private.
- **Cloud AI (Google Gemini):** For top performance, Google Gemini API offers faster responses and possibly better results.

A shared AI service layer lets the system switch between local and cloud models easily, based on user choice.

2.3 Core Features

The system's main features include:

- **Brainstorming Module:** An interactive chat to help develop thesis topics.
- **Timeline Generation:** AI creates detailed project timelines with tasks and time estimates based on user needs.
- **Task Management:** Integrates with Notion to create workspaces, sync tasks, and track progress automatically.
- **Academic Tools Suite:** Includes 18 specialized tools for grammar, web search, citation, PDF summaries, math solving, and academic integrity checks.

2.4 Ethical AI Monitoring

An ethics tracking system watches how AI is used, offers advice on academic integrity, and keeps usage logs to ensure AI helps responsibly.

3. Results

3.1 System Performance

The system successfully performs all its intended functions. Key performance points include:

- **Response Times:** AI-powered timeline generation takes 15-30 seconds.
- **Data Persistence:** Projects save and load successfully across different sessions.

3.2 Application Flow and User Experience

The system guides users through five main phases of thesis management:

3.2.1 Brainstorming Phase

This is where users start developing their thesis topics with AI help. It's an interactive chat that refines initial ideas into clear research topics.

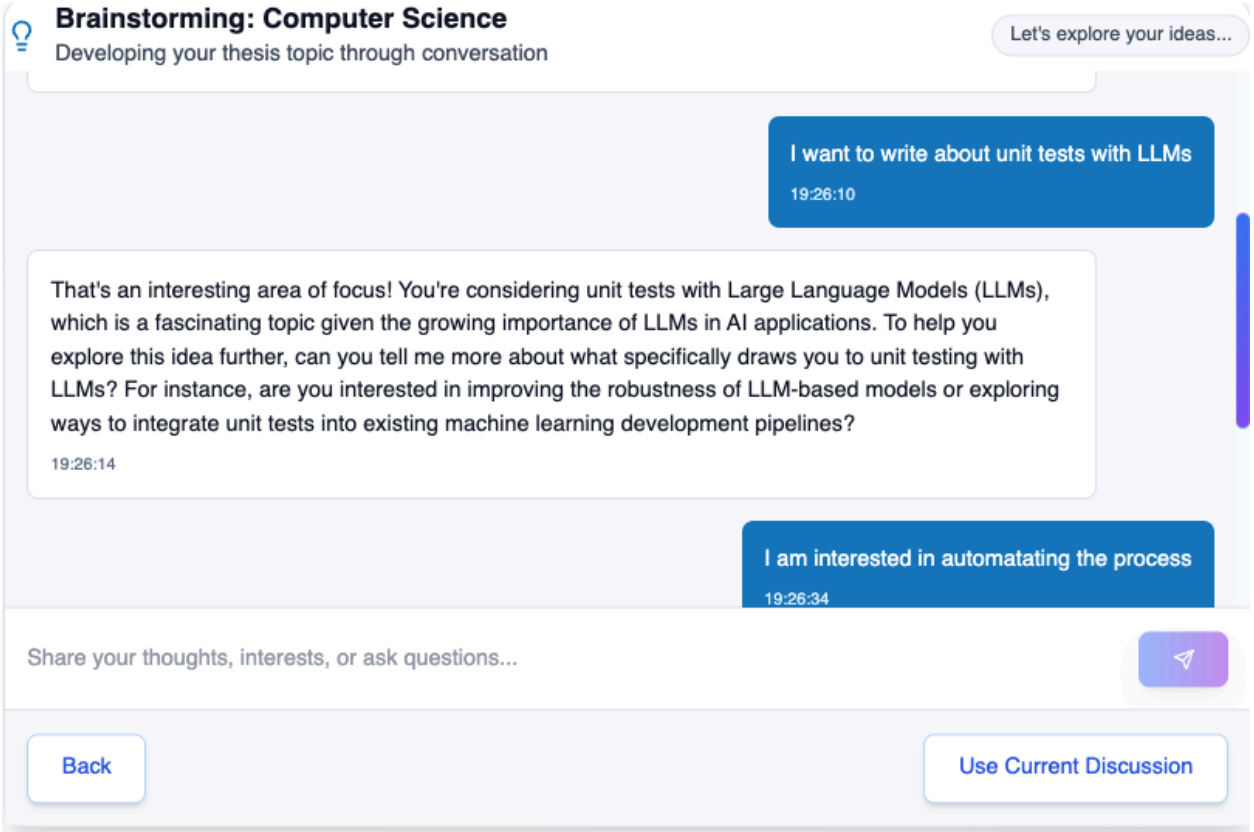

Figure 1: The brainstorming interface where users can interact with AI to develop their thesis topics.

3.2.2 Questionnaire Phase

Here, the system gathers detailed project needs to create an accurate timeline. This form collects important information about the thesis scope, deadlines, and user preferences.

Figure 2: The questionnaire that helps in creating a personalized timeline.

3.2.3 Timeline Creation

This phase uses AI to generate a full project plan from user requirements. It shows a visual timeline and connects with Notion for better project management.

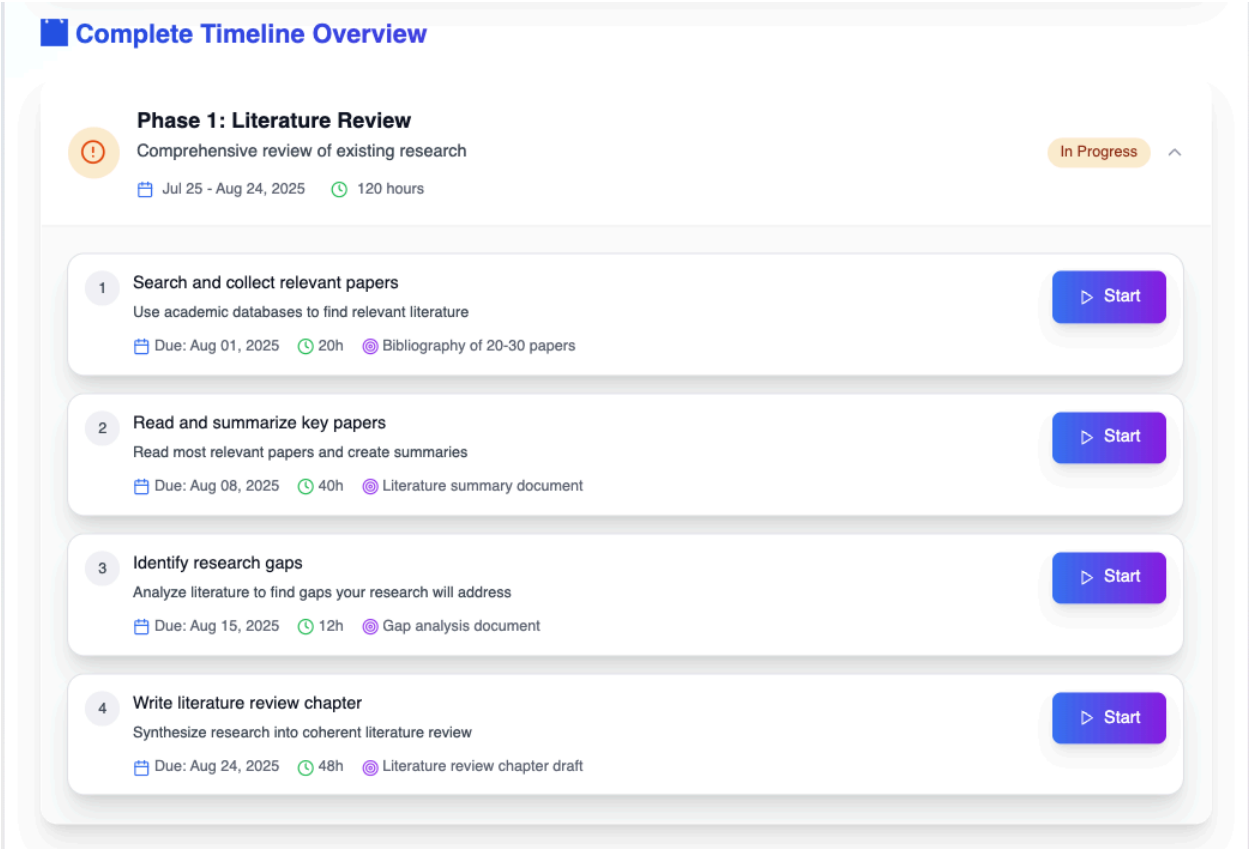

Figure 3: The timeline display shows a comprehensive project breakdown with phases, tasks, and milestones.

Add icon Add cover Add description

Thesis Tasks

Default view

New

Task	Complete...	Created	Description	Due Date	Estimated Hours	Order
Thesis Revision 2		July 13, 2025 8:12 PM	Final revisions based on advisor's feedback	July 20, 2025	18.75	23
Thesis Revision 1		July 13, 2025 8:12 PM	Revise the thesis based on self-review and	June 30, 2025	18.75	22
Draft Thesis (Full)		July 13, 2025 8:12 PM	Write a full draft of the thesis, including int	May 31, 2025	50	21
Results Write-up (Chapter 4)		July 13, 2025 8:12 PM	Write up the results chapter, integrating qu	March 15, 2025	25	20
Policy Analysis (Sweden)		July 13, 2025 8:12 PM	Analyze Swedish healthcare integration pol	February 15, 2025	12.5	19
Policy Analysis (Germany)		July 13, 2025 8:12 PM	Analyze German healthcare integration poli	February 15, 2025	12.5	18
Quantitative Data Analysis		July 13, 2025 8:12 PM	Analyze quantitative survey data using app	January 15, 2025	31.25	17
Qualitative Data Analysis		July 13, 2025 8:12 PM	Analyze qualitative interview data using the	January 15, 2025	31.25	16
Survey Data Collection		July 13, 2025 8:12 PM	Collect survey data from participants in bot	October 15, 2024	12.5	15
Conduct Interviews (Sweden)		July 13, 2025 8:12 PM	Conduct qualitative interviews in Sweden.	September 30, 2024	25	14
Conduct Interviews (Germany)		July 13, 2025 8:12 PM	Conduct qualitative interviews in Germany.	September 30, 2024	25	13
Participant Recruitment (Sweden)		July 13, 2025 8:12 PM	Recruit participants for interviews and surv	August 30, 2024	18.75	12
Participant Recruitment (Germany)		July 13, 2025 8:12 PM	Recruit participants for interviews and surv	August 30, 2024	18.75	11
Methodology Write-up (Chapter 3)		July 13, 2025 8:12 PM	Write a detailed description of the research	July 15, 2024	18.75	10
Survey Design & Pilot Testing		July 13, 2025 8:12 PM	Design the survey, pilot test it, and refine b	June 28, 2024	12.5	9
Ethical Review Application		July 13, 2025 8:12 PM	Prepare and submit the ethical review appli	June 14, 2024	6.25	8
Qualitative Interview Protocol Design		July 13, 2025 8:12 PM	Design interview protocols for migrant pati	June 7, 2024	6.25	7

Figure 4: The automatically created Notion workspace includes a main project page, task database, milestone tracking, and progress dashboard.

3.2.4 Email Notifications

The system sends automatic emails to track progress and offer motivation throughout the thesis project. This helps users stay engaged and informed.

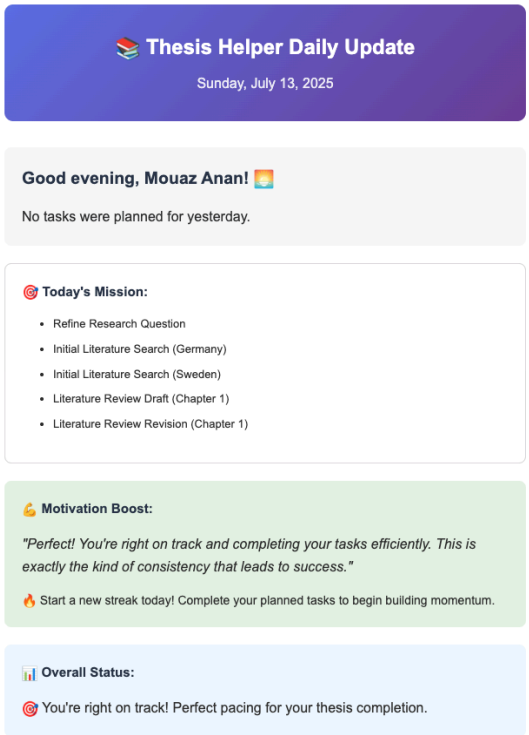

Figure 5: The email notification with project progress, upcoming tasks, and motivational content.

3.2.5 Task Execution with Specialized Tools

In this phase, users access 18 specialized academic tools through an integrated chat. This helps them work on specific tasks with AI assistance.

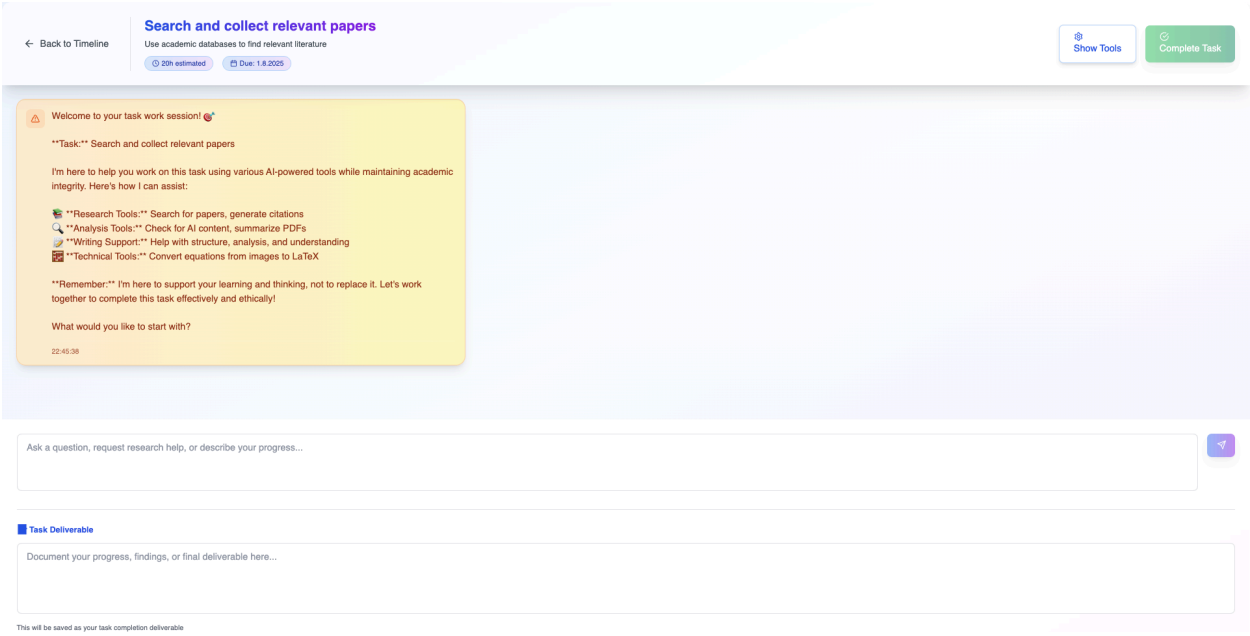

Figure 6: The task chat interface shows a dedicated workspace for working on specific thesis tasks.

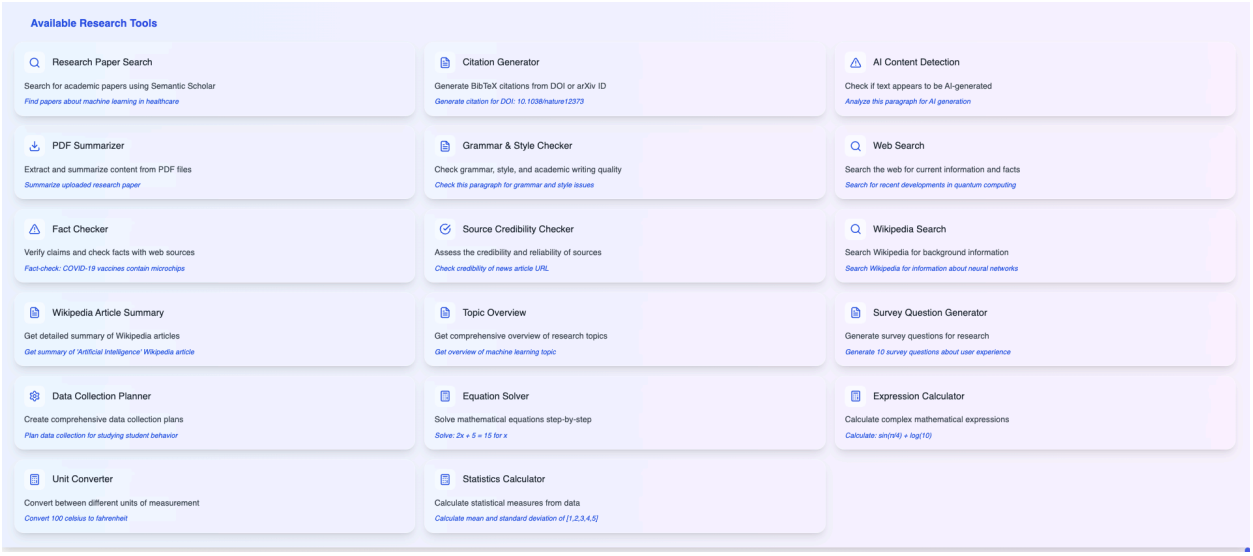

Figure 7: The academic tools panel displays the 18 specialized tools available to users. Each tool provides specific functionality for academic research and writing tasks.

3.3 Integration & AI Performance

External connections (Notion, Gmail, academic APIs like Semantic Scholar) work reliably. Both AI models effectively generate topics, create timelines, and assist with tasks appropriately.

3.4 Ethics and Academic Integrity

The ethics monitoring system successfully tracks AI use and guides users on academic integrity. Usage logs show that AI helps students without replacing their own work.

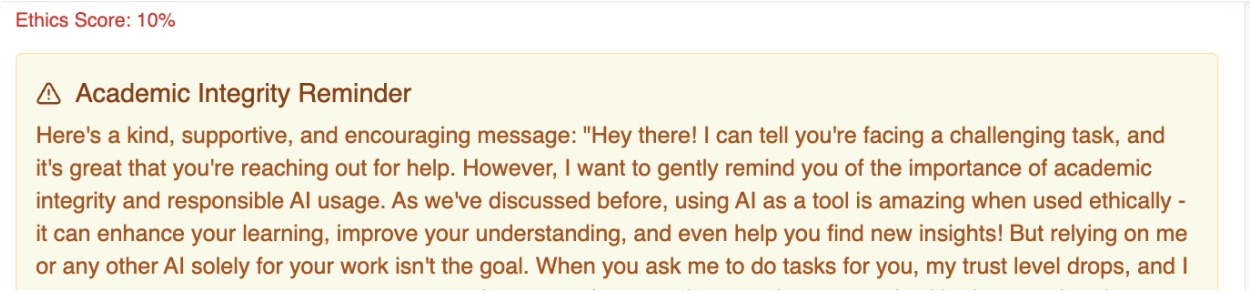

Figure 8: Ethics integrity warning

4. Discussion

4.1 Project Management Benefits

Notion integration greatly improves academic planning by providing professional project management features. Automated workspace creation makes it easy for students to use industry-standard tools.

4.2 Ethical Considerations

The ethics monitoring system handles a key concern in AI-assisted academic work. By tracking usage and giving guidance, it helps maintain academic integrity while allowing helpful AI use.

4.3 Limitations & Future Work

Current limits include reliance on external cloud AI services, and the need for ongoing maintenance as academic APIs change. Future improvements could include offline AI features, integration with school learning systems, and more language support.

Conclusions

This work successfully built a comprehensive AI-powered thesis management system that solves real problems in academic project management. The system effectively combines AI help with practical project management tools, all while maintaining academic integrity through built-in ethics monitoring.